PROJECT REPORT

# Brain Tumor MRI Classification

## with a Hybrid CNN–LSTM–PCA–SVM Framework

Five-Backbone Comparison, Explainability Analysis, Dataset-Integrity Audit, and Base-Paper Reproduction Verification

Project report submitted in fulfilment of the [project requirement]


*Submitted by*

**[Student name]**

Register No.: [register number]

[Programme / degree]

[University / institution]

*Under the guidance of*

**[Supervisor name and title]**

Department: [department]

Academic Year: [year]


*Research and educational prototype — not a clinically validated diagnostic system*

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# Chapter 1 · Abstract and Executive Summary

Magnetic resonance imaging (MRI) is one of the principal tools used to assess suspected brain tumors, and interpreting it demands specialist expertise. Deep-learning classifiers can reach very high accuracy on public MRI datasets, but a high score alone does not show that a model is reliable, that it is looking at the right part of the image, or that its evaluation was independent of its training data. This project set out to build a four-class MRI classifier and then, deliberately, to test how far its apparent performance could be trusted.

The system assigns a single two-dimensional brain MRI image to one of four categories: glioma, meningioma, pituitary tumor, or no tumor. It is a hybrid of deep and classical machine learning. A pretrained convolutional neural network (CNN) and a long short-term memory (LSTM) layer learn a 128-dimensional representation of each image; this is standardized, reduced to 64 dimensions by principal component analysis (PCA) and classified by a linear support vector machine (SVM). Five CNN backbones — VGG16, VGG19, ResNet50, ResNet152, and EfficientNet-B3 — are compared inside this common framework as independent alternatives. They are not an ensemble.

The design is derived from Bérété, Echtioui, Sellami and Ben Hmida, *Hybrid CNN+LSTM+SVM model for advanced brain tumor classification*, *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, article 43, 2026 (DOI 10.1007/s13721-025-00718-8) [1]. The present work is a **paper-inspired partial reproduction**: the architecture family, dataset, class scheme, learning rate, dropout, PCA size, and linear SVM regularization are the same as in the paper; the framework (PyTorch instead of TensorFlow), the optimizer coverage (RMSprop only, not the paper's five-backbone × four-optimizer matrix), the batch size (8 versus 32), the epoch budget (10 with early stopping, 30 for the two VGG variants), the freezing policy (VGG16 and VGG19 frozen after a training collapse), and the SVM probability setting (*probability=False* here versus *probability=True* in the paper) differ. Chapter 8 and Appendix A.6 lay these differences out one row at a time.

Technically the work is a Python pipeline built on PyTorch [13] and scikit-learn [14], with a Streamlit [15] research application on top. It was developed on Windows 11 with WSL2 Ubuntu and an NVIDIA RTX 5070 Laptop GPU. An initial TensorFlow implementation was abandoned because of GPU-architecture incompatibility — an environment problem, not a scientific judgement.

The study evolved through three questions. The first concerned classification and backbone comparison, and produced roughly 93–98% test accuracy. The second concerned explainability: Grad-CAM on the neural head and occlusion sensitivity on the complete pipeline showed that correct predictions often coincided with attention to skull margins, facial anatomy or image borders rather than to the visible lesion. The third concerned data integrity. A dedicated audit found material image-level duplicate and near-duplicate overlap across the dataset's supplied partitions.

A post-hoc leakage-corrected evaluation removed the implicated test images and re-scored the existing models without any retraining. It produced only small changes. On the 1,061-image material-clean test subset, EfficientNet-B3, ResNet152 and ResNet50 reached 98.680%, 98.398% and 98.303% accuracy, and VGG16 and VGG19 reached 94.816% and 93.025%. The differences among the top three are a few images and are not claimed to be statistically significant. The experimental phase was then locked. Important limitations remain: the models were developed on a split later found to contain duplication, no patient identifiers exist, no external dataset was used, and a single seed and split were run.

The base paper's own headline claim is that *ResNet152+LSTM+SVM (RMSprop) reaches 99% accuracy, 99% precision, 99% recall, and 99% F1* on the same Kaggle dataset [1, Table 3]. This project does not reproduce that 99% headline: our ResNet152 reaches 98.40% on the original test set and 98.398% on the material-clean subset — roughly 0.6 pp below the paper — and neither of the two most divergent backbones (VGG19 at 93.025% versus the paper's 96%) matches the paper exactly. Chapter 8 discusses the likely reasons: different batch size, different epoch budget, and the paper's unspecified freezing policy.

**Scope statement.** This is a research and educational prototype. It is not a clinically validated diagnostic system, does not segment tumors, and must not be used for medical decisions. Patient-level independence of the evaluation cannot be established. The report documents what was built, how the design evolved, the problems that were solved, the base paper's own methodology and how faithfully it was reproduced, and the honest limitations of the result.

## Project at a glance

| Aspect | Final state |
|---|---|
| Task | Four-class single-image brain MRI classification (glioma, meningioma, notumor, pituitary) |
| Dataset | Masoud Nickparvar Kaggle Brain Tumor MRI dataset [2]: 7,023 images (Training 5,712; Testing 1,311) |
| Pipeline | CNN → spatial sequence → LSTM(256) → BatchNorm → Dropout → Dense 128 → StandardScaler → PCA(64) → linear SVM |
| Backbones | VGG16, VGG19 (frozen); ResNet50, ResNet152, EfficientNet-B3 (fine-tuned) — independent alternatives |
| Primary final evaluation | 1,061-image material-clean test subset; highest observed accuracy EfficientNet-B3, 98.680% |
| Explainability | Grad-CAM (neural head) and occlusion (final SVM); formal evaluation on ResNet152 only |
| Integrity work | Image-level duplicate audit and post-hoc leakage-corrected evaluation |
| Base-paper positioning | Paper-inspired partial reproduction of Bérété et al. 2026 [1]; 3 project extensions (Grad-CAM, occlusion, integrity audit) |
| Interface | Streamlit research application |
| Status | Experimental phase complete and locked; documentation and submission phase |

## Key results (primary leakage-corrected evaluation)

| Backbone | Corrected test n | Accuracy | Macro F1 |
|---|---|---|---|
| EfficientNet-B3 | 1,061 | 98.680% | 98.511% |
| ResNet152 | 1,061 | 98.398% | 98.315% |
| ResNet50 | 1,061 | 98.303% | 98.134% |
| VGG16 | 1,061 | 94.816% | 94.302% |
| VGG19 | 1,061 | 93.025% | 92.415% |

*Results are for the 1,061-image material-clean test subset (one seed and split). EfficientNet-B3 has the highest observed accuracy; the differences among the top three are a few images and are not claimed to be statistically significant. The original 1,311-image results are reported in Chapter 6 for provenance.*

# Chapter 2 · Project Roadmap and Development Timeline

The project was not designed as a finished study on day one; it grew question by question, with each step responding to the result of the step before. The roadmap below presents that evolution as a sequence of deliberate milestones rather than a diary — each step solved a concrete limitation of the one before it.

### Development milestones

| # | Milestone | Solved |
|---|---|---|
| 1 | Selection of the base paper: Bérété et al. 2026, *Hybrid CNN+LSTM+SVM*, five CNN backbones | Provides a recent, comparable hybrid design |
| 2 | TensorFlow → PyTorch migration | RTX 50-series (sm_120) PTX incompatibility |
| 3 | Incremental build; first ResNet152 pipeline end-to-end | Shape tests; proven CNN–LSTM–PCA–SVM baseline |
| 4 | Generic five-backbone framework | One shared code path for VGG16/VGG19/ResNet50/ResNet152/EfficientNet-B3 |
| 5 | VGG collapse → frozen VGG backbones | Stable transfer learning; policy asymmetry documented |
| 6 | Original five-backbone results (~93–98% on 1,311-image test) | Answered the classification question |
| 7 | Streamlit research application | Made the pipelines usable and inspectable |
| 8 | Grad-CAM on the neural head | Started the explainability question |
| 9 | Full-pipeline final-SVM occlusion | Interrogates the deployed classifier that Grad-CAM cannot |
| 10 | Duplicate / leakage audit → **MATERIAL_DUPLICATION** | Revealed 562 material cross-split pairs |
| 11 | Post-hoc leakage-corrected evaluation (no retraining) | Quantified the effect; comparison preserved |
| 12 | Final experimental lock | Documentation, report and viva phase |

The project began with the selection of the Bérété et al. 2026 hybrid CNN+LSTM+SVM paper [1], which offered five comparable CNN backbones and a modern architecture to implement. An initial TensorFlow implementation failed on real GPU workloads because the RTX 50-series (Blackwell, *sm_120*) architecture was not supported by the toolchain in use, and the project moved to PyTorch. The architecture was then assembled and tested one piece at a time, a first ResNet152 pipeline was proven end to end, and it was generalized into a five-backbone framework.

During this generalization, fully fine-tuning VGG16 collapsed to near single-class prediction; freezing the VGG convolutional backbones stabilized training. The five original results were strong, and a Streamlit application was added to make the pipelines usable. High accuracy then raised a different question — *what is the model actually looking at?* — and Grad-CAM was added. Because Grad-CAM can only explain the differentiable neural head and not the deployed SVM, a second, gradient-free method (occlusion) was built to interrogate the full pipeline. Finally, before the project was frozen, a duplicate audit was run; it found material overlap between partitions, and a post-hoc leakage-corrected evaluation showed that the results were largely stable.

## 2.1 How the research questions evolved

The most important thread in the work is the way each result raised the next question. The project moved from *classification performance*, to *model comparison*, to *explainability*, to *dataset integrity and leakage*, to *faithfulness to the base paper*. What began as "I trained five models and obtained about 98%" became a study of performance, interpretability, data integrity, and reproduction discipline together.

| # | Question | Answer |
|---|---|---|
| Q1 | Can the hybrid framework classify four MRI classes accurately? | Yes at dataset level (≈98% for the top three backbones) |

| # | Question | Answer |
|---|---|---|
| Q2 | How do five CNN backbones compare under one common pipeline? | Top three within a few images; frozen VGG models lower |
| Q3 | Does high accuracy match plausible model attention? | Not consistently: peripheral, off-lesion maps were common |
| Q4 | Does Grad-CAM explain the deployed SVM decision? | No — it explains the neural head only |
| Q5 | Does full-pipeline SVM occlusion localize lesions better? | Sometimes, but not consistently |
| Q6 | Could duplicate leakage inflate the accuracy? | Material duplicate overlap existed (audit) |
| Q7 | Does removing contaminated test images collapse performance? | No — corrected results remain very similar |
| Q8 | Do the results match the base paper's headline? | No — ~0.6 pp below on ResNet152, ~3 pp below on VGG19 |

## 2.2 Timeline and evidence

Dates before 20 September 2026 come from project notes and are approximate; the entries for 20 September come from timestamps stored in the repository's CSV and JSON outputs. Git commit history was not used.

| When | Event | Source |
|---|---|---|
| 16 July 2026 | Early exploration of classical MRI-classification papers and pipelines | Project notes |
| 17 August 2026 | Bérété et al. 2026 hybrid CNN+LSTM+SVM paper chosen as base [1] | Project notes |
| 28 August 2026 | TensorFlow GPU failure; move to PyTorch | Project notes; environment freeze (Keras nightly of that date) |
| Late Aug – early Sep 2026 | Split saved; architecture built step by step; first ResNet152 pipeline | Project notes (approximate) |
| 6–12 September 2026 | Generic CNN–LSTM framework; VGG16 collapse and freeze; five-backbone results | Project notes |
| From about 12 September | Streamlit application; Grad-CAM and occlusion implemented | Project notes |
| 20 Sep 2026, 11:31–12:06 UTC | Grad-CAM ratings recorded (20 cases) | Grad-CAM CSV timestamps |
| 20 Sep 2026, from 12:43 UTC | Occlusion comparison ratings recorded (11 cases) | Occlusion CSV timestamps |
| 20 Sep 2026, 17:53 UTC | Duplicate audit run: MATERIAL_DUPLICATION | Audit JSON |
| After the audit | Leakage-corrected evaluation; experimental lock | Project record |
| 21 September 2026 | Base-paper verification against the primary source; traceability matrix and final assessment appended | This report, Appendix A.6 and A.7 |

# Chapter 3 · System Architecture and End-to-End Data Flow

This chapter describes how the parts of the system fit together and how data move through them. Five views are given: the layered architecture, the model pipeline with tensor shapes, the data partitions, the two explanation paths, and the integrity-audit flow. A final section traces what runs when one image is submitted to the application.

## 3.1 High-level architecture

The system has four layers. The **data layer** holds the dataset, the saved train/validation split and the preprocessing transforms. The **model-development layer** trains one CNN–LSTM per backbone, extracts deterministic 128-dimensional features from the best checkpoint, and fits the classical stage (StandardScaler, PCA, linear SVM) on training data only. The **analysis layer** examines the trained system: Grad-CAM explains the neural head, occlusion explains the final SVM, and the integrity audit examines the data and the evaluation. The **application layer** is a Streamlit interface that loads the saved artifacts for any of the five backbones.

| Layer | Contents |
|---|---|
| Data | Kaggle Brain Tumor MRI (7,023 images, 4 classes) → seed-42 stratified split (4,569 / 1,143 / 1,311) → preprocessing (224×224, ImageNet norm.) |
| Model development | Per-backbone CNN–LSTM training (5 backbones, RMSprop) → deterministic 128-D feature extraction → train-only StandardScaler → PCA(64) → linear SVM (C = 1.0) |
| Analysis | Grad-CAM on the neural head; final-SVM occlusion sensitivity; cross-split hashing/perceptual audit; leakage-corrected re-scoring |
| Application | Streamlit app (app.py, gui/) — loads experiments/<backbone>/ artifacts; SVM prediction, scores, metrics, Grad-CAM, optional occlusion |

## 3.2 Model pipeline and tensor shapes

Each backbone defines one complete pipeline. A preprocessed image passes through a pretrained CNN with its classification layers removed, so that it emits a spatial feature map. The feature map is rearranged into a sequence, read by a single-layer LSTM, and compressed by batch normalization, dropout and a dense layer into the 128-dimensional deep feature. A temporary linear "neural head" is attached during training; after training the feature is standardized, projected by PCA to 64 dimensions and classified by the linear SVM, which produces the system's final prediction.

| Stage | Shape (ResNet152, batch B) | Notes |
|---|---|---|
| Input | [B, 3, 224, 224] | RGB, ImageNet-normalized |
| CNN feature map | [B, 2048, 7, 7] | 512 channels for VGG, 1,536 for EfficientNet-B3; same 7×7 grid |
| Spatial sequence | [B, 49, 2048] | Each grid cell becomes one sequence element |
| LSTM output | [B, 49, 256] | One layer, hidden size 256, batch-first |
| Last position | [B, 256] | Output after the final (49th) element |
| Deep feature | [B, 128] | BatchNorm1d → Dropout(0.3) → Linear(256→128) → ReLU |
| Neural head (training only) | [B, 4] | Linear(128→4), used with cross-entropy loss |
| After StandardScaler / PCA | [B, 128] / [B, 64] | Both fitted on training features only |

**The LSTM does not see time.** The input is a single 2D MRI. The LSTM is not operating over time, sequential scans, patient history or MRI slices. It reads an artificial, raster-ordered sequence of the CNN's spatial positions (row by row, left to right, top to bottom). This lets the architecture follow the hybrid CNN–LSTM idea for single images. The base paper describes the same LSTM stage as capturing *"sequential dependencies across the feature maps"* [1, §3.2] — spatial rather than temporal — but the raster order is arbitrary and is a methodological limitation.

## 3.3 Data partitions

The dataset supplies a *Training* directory (5,712 images) and a *Testing* directory (1,311 images) but no validation set. A validation set was created from *Training* with a stratified 80/20 split and random seed 42, giving 4,569 training and 1,143 validation images; the split indices are saved to disk so that all five backbones use exactly the same images. The 1,311-image directory is called the **dataset-provided held-out test set**: it was not used to fit any model parameter, but it was used for evaluation and for sampling the explainability cases, so it is never described as "untouched".

| Partition | Images | Role |
|---|---|---|
| Train | 4,569 | Fits CNN–LSTM, scaler, PCA, SVM |
| Validation | 1,143 | Checkpoint selection, early stopping |
| Held-out test (dataset-provided) | 1,311 | Final evaluation; explainability sampling |

The class supports are as reported in the base paper for the same dataset [1, §3.1]: Training has glioma 1,321; meningioma 1,339; notumor 1,595; pituitary 1,457. Testing has glioma 300; meningioma 306; notumor 405; pituitary 300. A key finding of the audit (§6.3) is that the separately supplied Training and Testing folders do not, in fact, guarantee independent images.

## 3.4 Explainability data flow

Two explanation paths were built because the system contains two different classification heads. The CNN–LSTM trunk feeds a differentiable neural head, which Grad-CAM can explain through gradients. The deployed decision, however, is made by the scikit-learn PCA–SVM, and gradients cannot pass through it. Occlusion sensitivity therefore masks patches of the input and re-runs the *entire* pipeline, measuring the change in the SVM's decision value for the predicted class. Neither method produces a tumor segmentation.

| Method | Explains | How |
|---|---|---|
| Grad-CAM | Neural head (128 → 4 linear layer) | Back-propagates the class score through the CNN; averages positive contributions per activation map |
| Occlusion sensitivity | Deployed SVM (via full CNN–LSTM–scaler–PCA–SVM path) | Masks 32×32 patches on the input; re-runs the whole pipeline; records drop in the SVM decision value and count of prediction flips |

## 3.5 Integrity audit and correction flow

The integrity workflow hashes every image, compares the three partitions with each other, and classifies matches as exact, high-confidence near-duplicate or review-only. The verdict was **MATERIAL_DUPLICATION**. Rather than discarding the work, a post-hoc correction then loaded each experiment's saved test PCA features, labels and SVM, excluded the flagged test images under three predefined policies, and re-scored. Nothing was retrained or refitted.

## 3.6 What runs on a single prediction

A recurring design principle is that the final answer always comes from the SVM, while the neural head and explanation maps are supporting views. Tracing one uploaded image through the application shows where each component is used.

| Step | Component | What happens |
|---|---|---|
| 1 | Evaluation transform | Image converted to RGB, resized to 224×224, ImageNet-normalized (the same deterministic transform used for saved features) |
| 2 | CNN–LSTM (one forward pass) | Produces the neural-head logits and the 128-D deep feature |
| 3 | StandardScaler → PCA → SVM | Saved objects are applied (nothing is refitted); the SVM gives the final class and its decision values |
| 4 | Score display | Neural softmax scores and SVM "relative class scores" (softmax of decision values, not probabilities) |
| 5 | Grad-CAM (extra backward pass) | Heat map for the neural head's predicted class; one layer for VGG/EfficientNet, layers 3 and 4 plus fusion for ResNet |
| 6 | SVM occlusion (optional) | 169 masked copies of the image run through the whole pipeline in small batches; slower and coarse |

# Chapter 4 · Core System Modules and Logic

The project is organized into focused modules, each responsible for one concern. The table below maps the principal modules to their purpose and implementation.

| Module | Purpose | Main technology / file |
|---|---|---|
| Data pipeline | Transforms, saved-split loading, datasets and loaders | PyTorch / torchvision — src/data_pipeline_torch.py |
| Split creation | Stratified 80/20 split of Training; saves indices | scikit-learn — src/save_split.py |
| Dataset inspection | Readability, image modes and sizes | Pillow — src/check_dataset.py |
| CNN–LSTM model | Backbone, sequence conversion, LSTM, dense feature, neural head | PyTorch — model_cnn_lstm.py |
| Trainer | Training/validation loop, checkpointing, early stopping, resume | actual_training_scripts/train_backbone.py |
| Evaluator | Deterministic features, scaler, PCA, SVM, metrics | actual_training_scripts/evaluate_backbone.py |
| Grad-CAM | Neural-head explanation | gui/gradcam.py |
| Occlusion | Full-pipeline final-SVM explanation | gui/occlusion.py |
| Inference engine | Single-image inference; artifact validation | gui/inference_engine.py, model_factory.py, model_registry.py, results_loader.py |
| Application | Streamlit interface (four tabs) | app.py |
| Evaluation apps | Manual rating tools with fixed protocols and CSV export | gradcam_evaluation_app.py, occlusion_evaluation_app.py |
| Integrity audit | Hashing, cross-split comparison, audit reports | final_project_qc.py |
| Corrected evaluation | Post-hoc re-scoring without retraining | evaluate_leakage_clean_test.py |

## 4.1 Preprocessing and the data pipeline

All preprocessing is defined once, in *src/data_pipeline_torch.py*, and imported by training, feature extraction and the application, so that every stage sees identically prepared images. Two transforms exist: a training transform that includes random augmentation, and a deterministic evaluation transform.

| Step | Training transform | Evaluation transform |
|---|---|---|
| 1 | Convert to three-channel RGB | Convert to three-channel RGB |
| 2 | Resize directly to 224×224 | Resize directly to 224×224 |
| 3 | RandomHorizontalFlip (p = 0.5) | — |
| 4 | RandomVerticalFlip (p = 0.5) | — |
| 5 | ColorJitter: brightness 0.1, contrast 0.2, saturation 0.2, hue 0.1 | — |
| 6 | Convert to tensor | Convert to tensor |
| 7 | ImageNet normalization (mean 0.485, 0.456, 0.406; std 0.229, 0.224, 0.225) | Same normalization |

RGB conversion is necessary because the pretrained backbones expect three channels while many dataset images are single-channel. ImageNet normalization matches the inputs on which the pretrained weights were learned [8]. Augmentation is used only in training: it deliberately perturbs images to help the network generalize, but evaluation must be repeatable and must reflect unmodified data. The same deterministic transform is therefore used for validation, test images, the features that feed PCA and the SVM, and images uploaded to the application.

**Faithfulness to the base paper.** The paper's Algorithm 1 [1, §3.2] specifies horizontal and vertical flips, brightness with max delta 0.1, contrast in 0.8–1.2, saturation in 0.8–1.2, and hue with max delta 0.1. The project uses torchvision *ColorJitter(brightness=0.1, contrast=0.2, saturation=0.2, hue=0.1)*, which samples factors in *[1 – x, 1 + x]*. The effective ranges therefore coincide with the paper: 0.9–1.1 for brightness/hue and 0.8–1.2 for contrast/saturation.

Several things are deliberately absent: there is no skull stripping, no tumor segmentation, no aspect-ratio-preserving crop (the direct resize distorts non-square images), no MRI-specific intensity normalization beyond the ImageNet normalization, and no patient-level grouping. The classical handcrafted approaches considered in July 2026 (Otsu thresholding, K-means segmentation, GLCM or wavelet features) were never implemented, and the earlier TensorFlow input pipeline is legacy code.

## 4.2 The CNN–LSTM feature learner

*model_cnn_lstm.py* defines **CNNLSTM**, which builds one ImageNet-pretrained backbone [3][4][5] with its classification layers removed, so that it outputs a feature map of shape [B, C, H, W]. The map is permuted and reshaped to [B, H·W, C] by *cnn_to_sequence*. A single-layer LSTM [6] with hidden size 256 reads the sequence; its output at the last position is passed through batch normalization [9], dropout with probability 0.3 [10] and a linear layer from 256 to 128 units followed by ReLU. The resulting 128-dimensional vector is the deep feature. A linear layer from 128 to 4 units is attached as a temporary neural head so that the network can be trained end to end with cross-entropy loss.

### Why spatial sequencing, and what it costs

A CNN summarizes an image as a grid of feature vectors. For ResNet152 this is a 7×7 grid in which each of 49 cells carries 2,048 features. Flattening the grid gives a sequence of 49 elements that an LSTM can read. This is what makes the hybrid architecture implementable for single 2D images: it preserves the idea of a learned recurrent stage on top of CNN features and lets the network combine information across positions. It is also a genuine limitation. The raster order is arbitrary — a cell's vertical neighbor is seven steps away in the sequence while its horizontal neighbor is one step away — the model is not invariant to that order, and the project did not test an alternative ordering or a no-LSTM baseline. Whether the recurrent stage contributes anything beyond simple pooling of the feature map has therefore not been established.

## 4.3 Backbones and training policies

Exactly five backbones make up the study. *SUPPORTED_BACKBONES* in the model file lists these five and no others.

| Backbone | Character | Channels | ImageNet weights | Training policy |
|---|---|---|---|---|
| VGG16 [3] | 16 weight layers of stacked 3×3 convolutions | 512 | IMAGENET1K_V1 | Convolutional backbone frozen |
| VGG19 [3] | Deeper variant with 19 weight layers | 512 | IMAGENET1K_V1 | Convolutional backbone frozen |
| ResNet50 [4] | 50 layers with residual (skip) connections | 2,048 | IMAGENET1K_V2 | Trainable / fine-tuned |
| ResNet152 [4] | 152-layer residual network; base-paper-centered; formal explainability model | 2,048 | IMAGENET1K_V2 | Trainable / fine-tuned |
| EfficientNet-B3 [5] | Jointly scaled depth, width and resolution; designed for a larger native input than 224×224 | 1,536 | IMAGENET1K_V1 | Trainable / fine-tuned |

The plan was to fine-tune all five networks in the same way. When VGG16 was fully unfrozen, training became unstable: according to the project record, validation accuracy collapsed to about 27.9% and the

learned features became almost constant, so that PCA later produced invalid values. The figure is telling — the largest validation class, *notumor*, contains 319 of the 1,143 validation images, which is 27.9%, exactly what a network scores by always predicting one class. A diagnostic script (*diagnose_vgg16_collapse.py*) reported activation statistics for the untrained pretrained network, and according to the notes the pretrained activations were healthy. That located the fault in the training configuration rather than in the data, the class mapping or the pretrained weights. A single root cause was never isolated, and lower learning rates, partial unfreezing or gradient clipping were not studied.

The remedy was to freeze the pretrained VGG convolutional backbone (*requires_grad* = *False*) and train only the LSTM, batch normalization, dense layer and neural head. VGG16 then trained stably and VGG19 was given the same policy. The consequence must be stated plainly: the five models do not share identical optimization dynamics. The VGG networks act as fixed feature extractors with a trained read-out, whereas the others adapt their convolutional weights, so the accuracy gap between the two groups cannot be attributed to architecture alone. The correct wording is that each selected pretrained CNN backbone followed its model-specific freezing or fine-tuning configuration; the VGG16 and VGG19 convolutional backbones were frozen for training stability.

> **Difference from the base paper.** The base paper [1] does not describe any freezing or fine-tuning policy for the CNN backbones; its VGG16 result under RMSprop is 94% [1, Table 3], close to this project's frozen VGG16 at 94.816% (corrected). How the paper achieved stable VGG16 fine-tuning under the same 30-epoch RMSprop regime that triggered our collapse is not explained in the paper.

## 4.4 Training procedure

| Setting | Value |
|---|---|
| Loss / optimizer | CrossEntropyLoss on the neural head; RMSprop, learning rate 1e-4, trainable parameters only |
| Batch size / dropout | 8 / 0.3 (LSTM hidden 256, dense feature 128) |
| Random seed | 42 (Python, NumPy, PyTorch, CUDA); the training loader uses a seeded generator |
| Early stopping | Patience of 3 epochs without improvement |
| Checkpoint selection | Highest validation accuracy (strict improvement); checkpoint stores weights, optimizer state, history |
| Loaders | Training: shuffled, drop_last = True. Validation: not shuffled |
| Maximum epochs | VGG16 / VGG19: up to 30. ResNet50 / ResNet152 / EfficientNet-B3: up to 10 |

**Why drop_last.** The training set has 4,569 images and 4,569 = 571 × 8 + 1, so with a batch size of 8 the last batch of each epoch would contain a single image. Batch normalization computes its statistics from the batch during training and cannot do so from one sample, so the incomplete batch is discarded. Because the loader reshuffles every epoch, a different image is dropped each time and none is permanently excluded.

**Unequal budgets.** The VGG runs were extended to a 30-epoch target while the others used the trainer's default of 10 epochs, so the comparison is not perfectly epoch-matched; this is a stated limitation. Best epochs are stored as *best_epoch* = *epoch* + *1* and are therefore already 1-based: VGG16 16, VGG19 18, ResNet50 7, ResNet152 8, EfficientNet-B3 6.

> **Difference from the base paper.** The paper [1, Table 2] uses batch size 32, a fixed 30 epochs for every backbone, and no early-stopping (or the paper does not describe one). This project uses batch size 8 (for BatchNorm1d stability under limited GPU memory) and adaptive epoch budgets with patience-3 early stopping. The paper's optimizer study also covers Adam, SGD, AdaGrad and RMSprop; this project executed only the RMSprop branch.

## 4.5 Feature extraction and the final classifier

After training, the neural head is set aside and the network becomes a feature extractor. *evaluate_backbone.py* loads the best checkpoint, extracts deterministic 128-dimensional features for train, validation and test using the evaluation transform (no augmentation, no shuffling), and then fits, in order: a StandardScaler on the training features only; PCA with 64 components on the scaled training features only (retaining 99.18–99.45% of the training variance depending on the backbone); and a linear SVM [7], *SVC(kernel="linear", C=1.0, random_state=42, probability=False)*, on the training PCA features. Neither the PCA size nor C was tuned.

Fitting the scaler and PCA on training data only, and merely transforming validation and test data, prevents preprocessing leakage: statistics of the evaluation images never influence the model. This is separate from, and does not address, the duplicate-image leakage found later (Chapter 6). The deployed classifier is the SVM. The neural head exists to train the network and to provide a differentiable target for Grad-CAM; its prediction is displayed only for comparison.

Because the SVM is created with *probability=False*, it produces no probabilities. The application applies a softmax to the SVM's decision values purely to draw bars, and labels them *relative class scores*, never "probabilities". An older, root-level ResNet152 pipeline used *probability=True*; that legacy pipeline is superseded and must not be confused with the standardized experiments.

> **Difference from the base paper.** The base paper [1, §3.2] explicitly enables SVM probability estimation (*"Probability estimation was enabled to allow for probabilistic output"*). This project disables it and displays a softmax over *decision_function()* values as noncalibrated relative scores. The choice does not change the classification (the SVM decision boundary is identical); it only affects whether calibrated class probabilities are exposed.

## 4.6 Explainability modules

**Grad-CAM** [11] (*gui/gradcam.py*) back-propagates the score of a chosen class from the neural head to convolutional activations, weights each activation map by its average gradient, keeps the positive part, upsamples it to 224×224 and rescales it to [0, 1]. For ResNet backbones two layers are used — layer 3 (14×14, finer spatial detail) and layer 4 (7×7, more abstract features) — and a fused map averages the two equally; for VGG and EfficientNet a single last layer is used. Overlays use opacity 0.55. Because cuDNN recurrent kernels cannot run backward in evaluation mode, the LSTM alone is temporarily switched to training mode with cuDNN disabled for the explanation pass, while batch normalization and dropout stay in evaluation mode, so the result remains deterministic.

**Occlusion sensitivity** [12] (*gui/occlusion.py*) slides a 32×32 patch across the 224×224 input with stride 16, giving 13 × 13 = 169 occlusions. Each patch is replaced by zeros in normalized space (equal to the ImageNet channel means), and the masked image is passed through the complete pipeline. For each occlusion, the drop in the SVM decision value of the originally predicted class is recorded and prediction changes are counted. The drops are averaged over the patches that cover each pixel, negative values are discarded, and the map is divided by its own maximum. This last step is a key caveat: every occlusion map is normalized within its own image, so bright red means "largest effect in this image", not "large effect".

> **Project extensions vs. the base paper.** The base paper [1, §4.4] mentions Grad-CAM and Score-CAM only as future work in its Discussion and does not implement any explainability method. Grad-CAM on the neural head and occlusion sensitivity on the full pipeline are therefore **project extensions**, not part of the reproduced methodology.

## 4.7 Integrity-audit tooling

*final_project_qc.py* hashes every image and compares the three partitions with each other (Train ↔ Validation, Train ↔ Test, Validation ↔ Test) using five complementary measures. It does not enumerate duplicates lying entirely within a single partition.

| Measure | What it detects |
|---|---|
| File SHA-256 | Byte-identical files |
| Decoded-pixel SHA-256 | Identical pixels stored with different file encoding |
| 64-bit perceptual hash (pHash) | Visually similar images; compared by Hamming distance |
| 64-bit difference hash (dHash) | A second perceptual fingerprint based on neighboring-pixel gradients |
| Grayscale MAE and correlation | Pixel-level confirmation of similarity for perceptual candidates |

Perceptual candidates have pHash distance ≤ 4 and dHash distance ≤ 6. A candidate is a material near-duplicate only under a strict rule — (pHash ≤ 1, dHash ≤ 1, MAE ≤ 4 and correlation ≥ 0.9995) or (pHash ≤ 2, dHash ≤ 2, MAE ≤ 2 and correlation ≥ 0.9998). Similar candidates that do not meet it (MAE ≤ 12, correlation ≥ 0.995) are recorded as review-only, and contact sheets are generated for visual inspection.

> **Project extension vs. the base paper.** The base paper does not audit its dataset for cross-split image-level duplication. The audit here is a project extension performed on the paper's own partitioning.

## 4.8 User interface

The Streamlit [15] application (*app.py*) lets a user run any of the five finalized pipelines on an uploaded PNG, JPG, BMP or TIFF image (up to 25 MB; DICOM is not supported). It has four tabs. **Predict MRI** shows the final SVM prediction, the neural-head prediction, a classifier-agreement notice, SVM relative class scores, neural softmax scores, Grad-CAM with an adjustable overlay, and optional final-SVM occlusion. **Model performance** shows validation and test metrics, confusion matrices, per-class results and training curves. **Compare models** shows a table and chart across the five backbones. **Methodology** shows the inference path and artifact-readiness checks.

The application loads only the standardized artifacts under *experiments/<backbone>/* and never the legacy root-level files. Nothing is refitted from an uploaded image. The trained architecture is rebuilt without downloading ImageNet weights, and the loaded bundle is validated before use (class mapping, SVM class order, and scaler/PCA dimensions must agree). Only one backbone is held in memory per session. The interface carries a research warning, labels SVM outputs as relative scores, and describes explanation maps as coarse contribution maps rather than segmentations.

> **Finalization note.** The Model performance and Compare models tabs read each backbone's original *final_results.json*, which holds the pre-audit 1,311-image results, and the Methodology tab still describes all backbones as "fine-tuned", which is inaccurate for VGG16/VGG19. Both are software and documentation clean-up items; the authoritative final numbers are those of Chapter 6.

# Chapter 5 · Development Challenges and Testing-Driven Improvements

Most of the engineering value in this project lies in the problems that surfaced during development and the changes made to solve them. The table summarizes the major issues and their resolutions; only items actually encountered in the project are listed.

| Problem observed | Change introduced | Final benefit |
|---|---|---|
| TensorFlow failed on real GPU work (INVALID_PTX, RTX 50-series sm_120) | Migrated to PyTorch (CUDA 12.8 build) | Reliable GPU training and feature extraction |
| "LSTM for MRI" was ambiguous (temporal vs spatial) | Defined the spatial-token conversion and tested shapes | Accurate description; limitation documented |
| Fully unfrozen VGG16 collapsed (≈27.9% validation accuracy) | Froze VGG convolutional backbones (VGG19 likewise) | Stable transfer learning; policy asymmetry disclosed |
| BatchNorm1d cannot train on a one-sample last batch (4,569 = 571×8 + 1) | drop_last = True with per-epoch shuffling | Error-free training; no image permanently excluded |
| Random augmentation would make PCA/SVM features unstable | Separate deterministic evaluation transform; shuffle off for extraction | Reproducible features for PCA, SVM and the GUI |
| Legacy root-level and standardized artifacts could be mixed | Standardized experiments/<backbone>/ directories; GUI reads only those | No cross-contamination of checkpoints, scalers and SVMs |
| SVM scores could be mistaken for probabilities | probability=False; softmax of decision values labeled "relative class scores" | Honest score display |
| Grad-CAM cannot explain the external SVM; cuDNN LSTM backward fails in eval mode | Target the neural head; LSTM temporarily in training mode; add SVM occlusion | Two explanation paths with correct disclosure |
| A Grad-CAM layer produced a constant map (no spatial signal) | Display it blank instead of interpreting it | No false localization |
| Per-image normalization made tiny occlusion effects look dramatic | Report maximum score drops and flip counts with the maps | Caveat documented (Figure 6.5) |
| Duplicate images crossing partitions | Audit, then post-hoc corrected evaluation under three policies | Quantified impact; principal findings preserved |
| Best-epoch values misread as zero-based in early notes | Confirmed best_epoch = epoch + 1 in code and histories | Correct 1-based reporting |
| Base-paper reference incomplete (DOI / vol / article number) | Verified from primary source; completed as vol. 15 art. 43, DOI 10.1007/s13721-025-00718-8 (Appendix A.6) | Full traceable citation |
| Paper-vs-project methodology drift not consolidated in one place | Added base-paper reproduction traceability matrix (Appendix A.6) | Reviewer can check every deviation row by row |

## 5.1 Evolution of verification

Verification in this project grew with the system rather than being added at the end. The architecture was first assembled and checked piece by piece with small executable scripts, so that shape errors could not hide inside a monolithic model. Saving the split indices, checkpoints, fitted scaler, PCA and SVM for every backbone made later analysis possible without retraining. When the VGG16 run collapsed, a dedicated diagnostic separated a data or weight problem from a training problem. The two explainability studies were run through purpose-built applications with fixed protocols (seed, sample size, overlay opacity) and CSV export, so that ratings could be reproduced and audited; the occlusion comparison set was chosen before the occlusion results were viewed.

The last and most consequential check was the integrity audit, which tested an assumption that everything else had silently relied on — that the partitions were independent. The corrected evaluation then answered

how much that mattered. These are engineering and methodological checks; they are not a clinical evaluation.

In the final documentation stage, one further check was added: a systematic comparison of every relevant claim in this report against the base paper's own text and tables (see Chapter 8 and Appendix A.6). This surfaced the small language issues corrected in Chapters 3.2, 4.5, and Chapter 10, and provided the traceability matrix a viva panel needs.

## 5.2 Development and test artifacts

The smoke-test and diagnostic scripts below document how the pipeline was verified. They are executable scripts that print shapes and statistics rather than an assertion-based test suite, and some perform heavy work when imported.

| Script | What it verifies |
|---|---|
| test_resnet152_features.py | Convolutional output shape and the spatial-sequence conversion |
| test_lstm_section.py | LSTM mechanics on the sequence |
| test_post_lstm_layers.py | BatchNorm, dropout and dense path |
| test_complete_model.py | Complete forward pass in its different modes |
| test_training_step.py | Loss, gradients and one optimizer step |
| test_real_mri_batch.py | Real dataset batches and class mapping |
| actual_training_scripts/test_all_backbones.py | All five backbones instantiate and run |
| actual_training_scripts/test_best_checkpoint_features.py | Best-checkpoint loading and deterministic feature extraction |
| actual_training_scripts/diagnose_vgg16_collapse.py | Pre-training activation statistics (mean, std, range, non-zero fraction) |
| verify_gui_setup.py | GUI packages, source files and per-backbone artifacts |

# Chapter 6 · Experimental Results and Analysis

This chapter reports what was measured. Two sets of classification results appear. The **original** results use the full 1,311-image test set and are retained for provenance. The **leakage-corrected** results, obtained after the integrity audit, are the project's authoritative final classification results. A third subsection compares them numerically to the base paper's own results.

## 6.1 Training runs

| Backbone | Policy | Epochs (requested / completed) | Best epoch | Best val. acc. | Recorded time | ≈ s / epoch |
|---|---|---|---|---|---|---|
| VGG16 | Frozen | 30 / 19 | 16 | 94.58% | 155 s (resumed leg) | 17 |
| VGG19 | Frozen | 30 / 21 | 18 | 94.58% | 423 s | 20 |
| ResNet50 | Fine-tuned | 10 / 10 | 7 | 97.73% | 230 s | 23 |
| ResNet152 | Fine-tuned | 10 / 10 | 8 | 98.25% | 502 s | 50 |
| EfficientNet-B3 | Fine-tuned | 10 / 9 | 6 | 98.43% | 260 s | 29 |

*Best epochs are 1-based. VGG16 was trained for ten epochs and then resumed toward its 30-epoch target; it stopped three epochs after its best epoch. Times were recorded incidentally and are not controlled efficiency measurements. Validation accuracy is that of the neural head at each epoch.*

## 6.2 Original results (pre-audit)

| Backbone | NN val. acc. | SVM val. acc. | Original test acc. | Original macro F1 | Correct / 1,311 |
|---|---|---|---|---|---|
| VGG16 | 94.58% | 94.93% | 94.97% | 94.68% | 1,245 |
| VGG19 | 94.58% | 94.49% | 93.52% | 93.10% | 1,226 |
| ResNet50 | 97.73% | 97.55% | 98.09% | 98.01% | 1,286 |
| ResNet152 | 98.25% | 98.25% | 98.40% | 98.32% | 1,290 |
| EfficientNet-B3 | 98.43% | 98.51% | 98.47% | 98.41% | 1,291 |

These figures were first read as strong performance. The *notumor* class was consistently the easiest (recall above 99% for every backbone), and the largest errors were between glioma and meningioma: VGG19, for example, classified 37 of 300 gliomas as meningioma, whereas ResNet152 did so for 7. EfficientNet-B3 classified one more test image correctly than ResNet152 and five more than ResNet50 — not a basis for claiming statistical superiority, since the results come from one seed and one split. High accuracy, however, does not show *why* a model is correct, which motivated the explainability and integrity work that follows.

## 6.3 Dataset-integrity audit

The audit hashed all 7,023 images and compared the three partitions. Its verdict, recorded on 20 September 2026, was **MATERIAL_DUPLICATION**.

| Category | Cross-split pairs |
|---|---|
| Exact file or exact-pixel duplicates | 464 |
| High-confidence material near-duplicates | 98 |
| Total material pairs | 562 |
| Additional review-only candidates | 481 |

Pair counts overstate the number of affected images because one image can appear in several pairs. The number of *unique* images implicated by the material rule is 354 of 4,569 training images (≈7.75%), 122 of

1,143 validation images (≈10.67%) and 250 of 1,311 test images (≈19.07%). Even exact duplication alone implicated 216 unique test images (≈16.48%).

The Training and Testing folders were separately supplied by the dataset, yet the audit shows that some content was already identical or nearly identical across them. The precise statement is: *the supplied Training and Testing partitions contained pre-existing image-level duplicate and near-duplicate overlap.* This was not created by the project's own splitting code; train–validation overlap arises because duplicate content already existed within the original Training directory and the random split placed copies on both sides. The audit does not explain why the dataset has this structure. The flagged test images were concentrated in two classes: none of the glioma test images was flagged, while 109 of 306 meningioma, 136 of 405 notumor and 5 of 300 pituitary test images were. Patient identifiers are unavailable, so patient-level independence cannot be established.

> **Consequence for interpretation of the base paper.** The base paper [1] uses the same Kaggle Nickparvar partitioning without reporting any duplicate check. The 562 material cross-split pairs found here are a property of the paper's own dataset, so the paper's headline 99% is also subject to this leakage; the paper does not quantify it.

## 6.4 Leakage-corrected evaluation

Discovering duplication did not mean discarding the work. The first question was quantitative: how much do the duplicated test images actually affect the observed performance? A post-hoc evaluation (*evaluate_leakage_clean_test.py*) therefore re-scored the existing models after excluding flagged test images. No CNN was retrained, and the StandardScaler, PCA and SVM were not refitted. It used each experiment's saved test PCA features, saved labels and saved SVM. Three exclusion policies were defined before the corrected results were examined.

| Policy | What is excluded | Test n | glioma / mening. / notumor / pituitary | Role |
|---|---|---|---|---|
| Original | Nothing | 1,311 | 300 / 306 / 405 / 300 | Provenance |
| exact_clean | Exact file / pixel duplicates | 1,095 | 300 / 200 / 300 / 295 | Sensitivity |
| material_clean | Exact + high-confidence near-duplicates | 1,061 | 300 / 197 / 269 / 295 | Primary |
| conservative_clean | Material pairs + all review-only candidates | 926 | 300 / 197 / 134 / 295 | Sensitivity |

**Final primary results.** The material-clean table below is the authoritative final classification result of the project. *material_clean* was designated the primary policy in advance because it corresponds directly to the audit's prespecified definition of material duplication.

| Backbone | Corrected test n | Correct | Accuracy | Macro F1 | Δ accuracy vs original | Δ macro F1 |
|---|---|---|---|---|---|---|
| EfficientNet-B3 | 1,061 | 1,047 | 98.680% | 98.511% | +0.206 pp | +0.103 pp |
| ResNet152 | 1,061 | 1,044 | 98.398% | 98.315% | ≈ 0.000 pp | ≈ –0.001 pp |
| ResNet50 | 1,061 | 1,043 | 98.303% | 98.134% | +0.210 pp | +0.119 pp |
| VGG16 | 1,061 | 1,006 | 94.816% | 94.302% | –0.149 pp | –0.379 pp |
| VGG19 | 1,061 | 987 | 93.025% | 92.415% | –0.491 pp | –0.685 pp |

*(pp = percentage points.)* EfficientNet-B3 has the highest observed corrected accuracy. Its lead over ResNet152 is three images (1,047 versus 1,044 of 1,061) and over ResNet50 four images; because there is one seed and split, no confidence intervals and no paired test, it is not claimed to be statistically established. ResNet152 remains the formal explainability model because the explainability protocol was built around it.

Performance was essentially stable. Two models improved slightly, one was unchanged and two declined slightly, and no change exceeded half a percentage point; the ordering — three closely spaced

residual/EfficientNet models well above the two frozen VGG models — was preserved. This indicates that **the original high accuracy was not driven solely by the identified duplicated test images**. Sensitivity analyses agree in pattern:

| Backbone | exact_clean acc. (n = 1,095) | exact_clean macro F1 | conservative_clean acc. (n = 936) | conservative_clean macro F1 |
|---|---|---|---|---|
| EfficientNet-B3 | 98.630% | 98.450% | 98.488% | 98.419% |
| ResNet152 | 98.447% | 98.331% | 98.164% | 98.269% |
| ResNet50 | 98.356% | 98.163% | 98.056% | 97.996% |
| VGG16 | 94.977% | 94.377% | 94.060% | 94.031% |
| VGG19 | 93.242% | 92.497% | 92.009% | 92.187% |

The stricter policy costs the two VGG models somewhat more, but it also removes many notumor images (405 down to 134), which changes the class balance markedly; it is best read as a robustness check rather than a replacement estimate. Because the class mix changes under every policy, before/after differences also reflect composition and not only the removal of leaked images.

> **The limitation that remains.** The corrected evaluation removes *identified* contaminated images from the *test* set. The networks, however, were trained and their checkpoints selected on a train/validation split later shown to contain duplication, so validation accuracy and best-epoch choices may be optimistic. Patient identifiers are unavailable, near-duplicates below the audit thresholds may remain, the analysis is post-hoc, and no external dataset was used. The project therefore does **not** claim the results are "completely leakage-free"; the accurate description is **leakage-corrected at the identified test-image level**, and the results are not proof of patient-independent or external clinical generalization.

For the scope of this academic project, the experimental stage was locked without a full duplicate-group-aware retraining, because the corrected results were stable and the main comparison remained intact. Such a rerun would be methodologically stronger for publication or for stronger generalization claims and is treated as future work (Chapter 9).

## 6.5 Direct numeric comparison with the base paper

The base paper [1] reports the same five backbones on the same dataset with four optimizers. To keep the comparison controlled we restrict the paper's numbers to the RMSprop column (the optimizer this project used) and compare against both project result sets.

| Backbone | Paper (RMSprop) accuracy [1, Table 3] | Project test acc. | Project material_clean acc. | Δ (project − paper), corrected |
|---|---|---|---|---|
| VGG16 | 94% | 94.97% | 94.816% | +0.8 pp |
| VGG19 | 96% | 93.52% | 93.025% | –3.0 pp |
| ResNet50 | 99% | 98.09% | 98.303% | –0.7 pp |
| ResNet152 | 99% | 98.40% | 98.398% | –0.6 pp |
| EfficientNet-B3 | 98% | 98.47% | 98.680% | +0.7 pp |

Two observations. First, three of the five backbones (VGG16, ResNet50, ResNet152) match the paper within ≈1 pp, with the two residual networks landing consistently below the paper's 99% by about half a percentage point; EfficientNet-B3 actually exceeds the paper's 98% by ≈0.7 pp on the corrected set. Second, the largest gap — **–3 pp on VGG19** — matches the pattern that VGG has been the most fragile family in this project (a full collapse under unfrozen training, only partial recovery under freezing), and is consistent with the paper's own text noting that even VGG19 "struggle[s] with multi-class classification" [1, §4.3].

The paper's headline claim — **99% accuracy, precision, recall and F1 for ResNet152+LSTM+SVM under RMSprop** [1, Abstract; Table 3, Table 4] — is therefore **not reproduced** here at the same headline number. Given the project used a different batch size (8 vs 32), a different (shorter) epoch budget for the non-VGG backbones (10 with early stopping vs 30 fixed), and a frozen VGG policy that the paper does not

describe, this partial-reproduction outcome is credible rather than surprising.

## 6.6 Explainability results

The formal evaluations used **ResNet152 only** and must not be generalized to the other backbones. The Grad-CAM protocol sampled the test set with seed 42, five images per class (20 images: 15 tumor, 5 notumor), with layer 3, layer 4 and fused maps at opacity 0.55. Localization was rated manually as Good, Partial or Poor (not applicable for notumor). The ratings are subjective, come from a single rater, and used no radiologist review or expert lesion masks.

| Map (15 tumor images) | Good | Partial | Poor | Good + Partial | Rated most relevant |
|---|---|---|---|---|---|
| Layer 3 (14×14) | 1 | 9 | 5 | 10 / 15 | 9 / 15 |
| Layer 4 (7×7) | 3 | 2 | 10 | 5 / 15 | 6 / 15 |
| Fused (50 / 50) | 2 | 6 | 7 | 8 / 15 | 0 / 15 |

The neural head and the SVM agreed on all 20 images, and both were correct on 19. The one shared error was *pituitary/Te-pi_0078.jpg*, classified as glioma, for which all three maps were rated Poor. The central observation is that correct classification often coexisted with poor localization: maps were frequently dominated by the skull margin, facial or orbital anatomy, extracranial regions or image borders, and only occasionally overlapped the visible lesion. One notumor image produced a constant map for one layer, displayed blank; another (*Te-no_0216*) shows a conspicuous rounded bright region and was flagged for expert review as a possible label anomaly — it was not relabeled.

The occlusion comparison used 11 cases chosen from the Grad-CAM set before the occlusion results were viewed (9 tumor, 2 notumor). For the nine tumor cases, occlusion localization was rated Good 2, Partial 2, Poor 5. Grad-CAM was judged more anatomically relevant in 4 cases, occlusion in 3, both were predominantly off-lesion in 2, and in none were both clearly lesion-focused. Occlusion therefore did not solve the interpretability problem. Across 11 × 169 = 1,859 occlusions, 58 changed the predicted class (≈3.12%); the median per-image flip rate was 0%, and 5 of the 11 images had any flip. The misclassified *Te-pi_0078* was unusually unstable: 38 of its 169 occlusions (≈22.5%) changed the prediction, versus 20 of 1,690 (≈1.18%) across the ten correctly classified cases.

> **Normalization caveat.** Every occlusion heat map is rescaled to its own maximum. For four of the eleven images, masking any single patch changed the SVM decision value by less than 0.01, yet each of their heat maps is scaled to the same 0–1 range as those of images with effects several hundred times larger and looks equally vivid. Occlusion maps should therefore always be read with their numerical effect sizes and never compared visually across images.

### Leakage-aware view of the explainability samples

The audit later showed that two explainability images were material duplicates: *meningioma/Te-me_0012.jpg* (an exact-pixel duplicate of a training image) and *notumor/Te-no_0060.jpg* (an exact-file duplicate of a validation image). Both appear in the 20-case Grad-CAM protocol and the 11-case occlusion comparison. The original experiments and CSV files are preserved unchanged for provenance and nothing was re-run. For leakage-aware descriptive discussion the samples reduce to 18 Grad-CAM cases (14 tumor, 4 notumor) and 9 comparison cases (8 tumor, 1 notumor). Neither removed image had any prediction flip, so all 58 flips remain among the smaller set of occlusions and the per-image median flip rate becomes about 0.6% rather than 0%. Once *Te-me_0012* is removed, no layer-3 map is rated Good. The core interpretation is unchanged.

## 6.7 Summary of findings

- The hybrid CNN–LSTM–PCA–SVM framework classifies the four MRI classes accurately at the dataset level: 98.3–98.7% corrected accuracy for the top three backbones and about 93–95% for the two frozen VGG backbones.

• The top three backbones differ by a few images; no statistical superiority is claimed. The VGG gap partly reflects the frozen policy and the different epoch budgets.

• High accuracy did not correspond to consistently lesion-centered explanations: Grad-CAM and occlusion each frequently highlighted non-lesion regions. This indicates plausible shortcut cues but does not prove them.

• The supplied Training and Testing partitions contained pre-existing image-level duplicate and near-duplicate overlap. The base paper uses the same partitions and does not audit them.

• Removing the identified contaminated test images changed accuracy by less than half a percentage point for every backbone, preserving the comparison; the caveats about train/validation leakage and patient-level independence remain.

• The base paper's headline 99% ResNet152 result [1] was not reproduced at the same headline number: this project's ResNet152 lands ≈0.6 pp below on both original and corrected test sets. Three of five backbones fall within ≈1 pp of the paper; VGG19 is ≈3 pp below.

# Chapter 7 · System Merits, Demerits, and Operational Constraints

This chapter states the strengths and the limitations of the project honestly. The limitations are real and important; presenting them clearly is part of responsible research, not an admission of failure.

## 7.1 Merits

• **Standardized comparison:** the five backbones share one preprocessing pipeline, split, seed, optimizer and evaluation code, so differences are observable under a common design (with the caveats listed below).

• **Reproducible artifacts:** saved split indices, checkpoints, fitted scaler, PCA and SVM for every backbone, plus result JSON files and rating CSVs, allowed later analysis — including the corrected evaluation — without retraining.

• **Leakage control at the preprocessing stage:** the scaler, PCA and SVM are fitted on training data only.

• **Explicit separation** of the neural head and the deployed SVM, with an explanation method matched to each (Grad-CAM for the head, occlusion for the full pipeline).

• **Explainability with documented limits:** effect sizes and flip counts are reported alongside occlusion maps, and the per-image normalization caveat is stated.

• **Data-integrity audit:** five complementary similarity measures across all three partition pairs, with reviewable contact sheets, followed by a corrected evaluation that quantifies the effect without discarding the work.

• **Transparent reporting:** original results are preserved beside the corrected ones, top-model differences and remaining leakage caveats are stated rather than hidden, and the base-paper comparison is quantified in Chapter 6.5 and Appendix A.6.

• **Traceable base-paper reproduction:** Appendix A.6 lists every base-paper element and states whether it is implemented, deviated, or extended — a viva panel can inspect the entire reproduction commitment on one page.

## 7.2 Demerits and operational constraints

• **Dataset overlap:** the supplied Training and Testing partitions contain pre-existing duplicate and near-duplicate images, and the train/validation split contains duplicated content, so validation scores and checkpoint selection may be optimistic.

• **Post-hoc correction:** the corrected evaluation removes only identified test duplicates; it is not a group-aware retraining and does not establish independence.

• **No patient-level analysis:** no reliable patient identifiers exist, so patient-level independence cannot be established, and there is no external dataset.

• **Limited statistical evidence:** one main seed and split, no repeated runs, no confidence intervals and no significance tests; the top three models differ by a few images.

• **Unequal comparison:** VGG backbones are frozen while the others are fine-tuned, and epoch budgets differ (up to 30 versus up to 10).

• **Architectural limitation:** the LSTM reads artificially ordered spatial tokens, and no no-LSTM, no-PCA or CNN-only ablation was run, so the contribution of each component is unmeasured.

• **Preprocessing simplicity:** direct resize, no skull stripping and no segmentation, so the model can use skull, background and border information.

• **Small qualitative explainability study:** ResNet152 only, 20 and 11 fixed cases, single-rater subjective ratings, no expert masks, and two cases later found to be duplicates.

• **Uncalibrated scores and no clinical validation:** SVM outputs are relative class scores, not probabilities; the system is a research and educational prototype and must not be used for diagnosis.

• **Partial reproduction of the base paper:** RMSprop only, different batch size, VGG freezing not in the paper, *probability=False* versus paper's enabled probability estimation. The 99% headline is not reproduced.

# Chapter 8 · Comparative Engineering Analysis (base-paper verified)

The project is compared along four axes: against its own early legacy pipeline, across the five backbones, against the base paper it was inspired by, and against the state-of-the-art table published in the base paper itself. The intent is to show what changed and where the work sits — not to claim superiority.

## 8.1 Early legacy pipeline versus final framework

| Aspect | Early legacy pipeline | Final framework |
|---|---|---|
| Framework | TensorFlow attempt, then a root-level PyTorch ResNet152 pipeline | Generic PyTorch CNN–LSTM for five backbones |
| Scope | One backbone (ResNet152) | Five independent backbones |
| Artifacts | Root-level models/ and pca_features/ | Standardized experiments/<backbone>/ bundles |
| Backbone policy | End-to-end fine-tuning | Model-specific (VGG16/VGG19 frozen) |
| SVM setting | probability = True | probability = False; relative scores from decision values |
| Evaluation | Accuracy, macro/weighted metrics and confusion matrix printed to the console | Metrics saved as JSON per backbone (per-class reports, confusion matrices); corrected evaluation added |
| Explainability | None | Grad-CAM (neural head) and occlusion (final SVM) |
| Integrity checks | None | Duplicate audit and leakage-corrected evaluation |
| Interface | None | Streamlit research application |

The legacy ResNet152 run reached 97.64% test accuracy according to project notes, against 98.40% for the standardized ResNet152 on the same original test set. The 0.76-point gap between two runs of the same general design is larger than the spread among the top three standardized models. It is not a variance estimate, since the implementations differed, but it is a further reason for caution about small differences.

## 8.2 The five backbones compared

| Backbone | Policy | Best epoch | ≈ s / epoch | Corrected accuracy | Corrected macro F1 | Correct / 1,061 |
|---|---|---|---|---|---|---|
| VGG16 | Frozen | 16 | 17 | 94.816% | 94.302% | 1,006 |
| VGG19 | Frozen | 18 | 20 | 93.025% | 92.415% | 987 |
| ResNet50 | Fine-tuned | 7 | 23 | 98.303% | 98.134% | 1,043 |
| ResNet152 | Fine-tuned | 8 | 50 | 98.398% | 98.315% | 1,044 |
| EfficientNet-B3 | Fine-tuned | 6 | 29 | 98.680% | 98.511% | 1,047 |

The two frozen VGG models are the fastest per epoch and the least accurate; ResNet152 is the slowest per epoch; EfficientNet-B3 has the highest observed accuracy at roughly 29 s per epoch. These are single runs with incidental timings, so they describe what happened rather than establish a ranking of efficiency or accuracy. The gap between the VGG models and the others reflects both architecture and the frozen policy and different epoch budgets.

## 8.3 This project versus the base paper

The verification of the report's paper-claims against the primary source (Bérété et al. 2026 [1]) is summarized here; the full traceability matrix is in Appendix A.6.

| Aspect | Base paper [1] | This project |
|---|---|---|
| Publication | *Netw. Model. Anal. Health Inform. Bioinform.*, vol. 15, art. 43 (2026); DOI 10.1007/s13721-025-00718-8; published online 02 Feb 2026 | Paper-inspired partial reproduction |
| Task | Four-class MRI tumor classification | Same |
| Dataset | Kaggle Nickparvar, 7,023 images, 5,712 + 1,311 partitioning | Same |
| Class supports (test) | glioma 300, mening. 306, notumor 405, pituitary 300 | Same (before audit); post-audit reduced under three exclusion policies |
| Framework | TensorFlow (Alg. 1 uses TF image transformation functions) | PyTorch (RTX 50-series sm_120 forced the migration) |
| Backbones (5) | VGG16, VGG19, ResNet50, ResNet152, EfficientNetB3 [1, §3.2] | Same five |
| Augmentation | H/V flip; brightness ±0.1; contrast 0.8–1.2; sat 0.8–1.2; hue ±0.1 [1, Alg. 1] | torchvision ColorJitter(0.1, 0.2, 0.2, 0.1) — same effective ranges |
| Freezing policy | Not specified | VGG16 and VGG19 CNN backbones frozen (project addition) |
| LSTM description | "Sequential dependencies across the feature maps" [1, §3.2] | Spatial-position sequence from CNN feature map (same behaviour, clearer wording) |
| StandardScaler → PCA(64) → linear SVM | Yes [1, §3.2] | Same |
| SVM `probability` | True (enabled) [1, §3.2] | False; GUI uses decision_function() as relative scores |
| Optimizer study | 5 backbones × 4 optimizers (Adam, SGD, AdaGrad, RMSprop) [1, §4.2, Tab. 2] | RMSprop only |
| Batch size | 32 [1, Tab. 2] | 8 |
| Epoch budget | 30 fixed [1, Tab. 2] | 10 default with early stopping; VGGs extended to 30 |
| Learning rate | 1e-4 [1, Tab. 2] | Same |
| Dropout | 0.3 [1, Tab. 2] | Same |
| Explainability | None (Grad-CAM/Score-CAM mentioned only as future work [1, §4.4]) | Grad-CAM on the neural head; occlusion on the full pipeline |
| Duplicate / leakage audit | None | Cross-split hashing + pHash/dHash + MAE/correlation on all 7,023 images |
| Leakage-corrected evaluation | None | Three exclusion policies; primary *material_clean* n = 1,061 |
| Streamlit / GUI | None | Four-tab research application |
| Statistical rigour | Single-seed / single-split; near-identical 99% headlines (no CIs or significance tests reported) | Same (single-seed, single-split) |

The project makes no claim to match, exceed or explain the paper's reported accuracy, and the paper's figures are never presented as this project's results. Where deviations are deliberate (framework / batch / optimizer scope), they are recorded here and in Chapter 4. Where deviations are project-specific stabilizations (VGG freezing), they are recorded here and in Chapter 4.3. The quantitative gap between paper and project is set out in Chapter 6.5.

## 8.4 Positioning against the base paper's own SoTA table

The base paper concludes with a comparative table against several recent multi-class methods on the same dataset [1, Table 4]. For informational context only (this project makes no headline claim against those systems), the paper's reported accuracies are:

| Reference (as cited by the base paper) | Model | Reported acc. |
|---|---|---|
| Asif et al. 2023b | ResNet152V2 + Deep Dense Block (DDB) | 96.10% |
| Khaliki & Ba■arslan 2024 | VGG16 (best of transfer-learning study) | 98% |
| Bilal & Asif 2025 | MobileNet feature fusion (w/o attention) | 96.08% |
| Kantu et al. 2025 | SqueezeNet | 96.64% |
| Amin et al. 2025 | ADE_DieT (transformer) | 96.09% |
| Gupta et al. 2025 | FCBTC Model-3 (ResNet50 + DenseNet121) | 98.40% |
| Bérété et al. 2026 [1] | ResNet152+LSTM+SVM (RMSprop) — the base paper | 99% |
| This project | ResNet152+LSTM+SVM (RMSprop) — corrected, n = 1,061 | 98.398% |
| This project | EfficientNet-B3+LSTM+SVM (RMSprop) — corrected, n = 1,061 | 98.680% |

Two honest notes must accompany any such table. First, none of the systems listed above audits its dataset partitions for image-level duplication, so all quoted accuracies are on partitions that this project's audit has shown to contain 562 material cross-split pairs. The *relative* ordering is therefore more informative than any single headline percentage. Second, the project's numbers are single-seed / single-split, so a difference of a few tenths of a percentage point should not be read as method-level superiority.

## 8.5 Where the project sits

Hybrid pipelines that combine CNN features with a classical classifier are common for brain-tumor MRI, and Grad-CAM [11] and occlusion [12] are established interpretability methods. This project does not claim to outperform published systems, and it introduces no novel network architecture. Its emphasis is **methodological**: it separates the explanation target from the deployed classifier, reports effect sizes alongside occlusion maps, and audits the dataset partitions and quantifies the effect of leakage. Readers comparing accuracy figures reported on this public dataset should bear in mind that its supplied partitions contain image-level overlap (Section 6.3), a finding the base paper does not report.

# Chapter 9 · Project Outcomes, Skills Learned, and Future Recommendations

## 9.1 Final outcome

The project delivers a working, documented five-backbone CNN–LSTM–PCA–SVM system for four-class brain-MRI classification, a Streamlit research application, a formal Grad-CAM and occlusion analysis, a dataset-integrity audit, a leakage-corrected evaluation, and a fully traced comparison against the base paper. Its central lesson is that **performance, interpretability, data integrity, and reproduction faithfulness must be examined together**: the strong accuracy was real at the dataset level and survived correction, but it did not imply lesion-centered reasoning, could not be interpreted safely until the dataset partitions had been audited, and did not match the base paper's own 99% headline.

> **Final status.** The experimental/model-development phase is complete and locked. The primary final evaluation is the 1,061-image material-clean test subset: EfficientNet-B3 has the highest observed accuracy (98.680%), ResNet152 reached 98.398% and remains the formal explainability model, and ResNet50 reached 98.303%, with no claim of statistical superiority among them. No additional training is planned unless an actual technical error is discovered. The system remains a research and educational prototype and is not a clinically validated diagnostic system.

## 9.2 Skills learned

The project developed applied machine-learning and engineering skills: PyTorch model construction and transfer learning; diagnosing a GPU-toolchain failure and a training collapse; classical machine learning with scikit-learn and the discipline of fitting only on training data; explainability with Grad-CAM and occlusion, including their limits; building a Streamlit application around saved artifacts; data-integrity auditing with cryptographic and perceptual hashing; reproducible experiment management; and the scientific discipline of reporting limitations honestly and revising interpretations when new evidence arrived. The final documentation stage added one further skill: **reading a base paper against the actual implementation and articulating exactly what was reproduced, what was deviated from, and what was extended**.

## 9.3 Future recommendations

The following items would strengthen the methodology. They are future work, not unfinished requirements of the current, locked project.

- **Duplicate-group-aware experiment:** connect duplicate and near-duplicate images into groups, split whole groups, retrain and evaluate on a group-independent holdout; use patient or study identifiers for a patient-level split if they become available.
- **Complete the base-paper matrix:** run the Adam / SGD / AdaGrad branches to complete the 5 × 4 optimizer study the paper describes; match the paper's batch size 32 and 30-epoch fixed schedule where hardware allows.
- **External validation:** evaluate on an independent dataset to test generalization beyond this collection.
- **Statistical rigour:** repeated seeds, confidence intervals and paired significance tests to judge whether the small differences among the top models — and the small gaps to the base paper — are meaningful.
- **Ablations:** CNN-only, no-LSTM and no-PCA baselines and an alternative sequence order, to measure what each component contributes and whether the recurring lower-right Grad-CAM hotspot relates to the raster-ordered read-out.

• **Calibration and quantitative explainability:** calibrated probabilities (matching the paper's *probability=True*), and expert lesion masks with localization metrics in place of subjective ratings; a clinically validated segmentation would be required for any boundary claim.

• **Robustness and clinical readiness:** scanner and site robustness studies, DICOM handling and prospective evaluation before any clinical discussion.

# Chapter 10 · References

The base-paper reference [1] and dataset citation [2] are the project's own primary sources and are fully verified below. References [3]–[15] are standard sources for the methods and tools used.


| | |
|---|---|
| [1] | M. Bérété, A. Echtioui, L. Sellami, and A. A. Ben Hmida, "Hybrid CNN+LSTM+SVM model for advanced brain tumor classification," *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, art. 43, 2026. DOI: 10.1007/s13721-025-00718-8. Received 9 July 2025; Revised 6 December 2025; Accepted 9 December 2025; Published online 02 February 2026. |
| [2] | M. Nickparvar, *Brain Tumor MRI Dataset*, Kaggle. Available: https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset. Version, licence and download date: citation completion required at submission. |
| [3] | K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015. |
| [4] | K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. |
| [5] | M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114. |
| [6] | S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997. |
| [7] | C. Cortes and V. Vapnik, "Support-Vector Networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995. |
| [8] | J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A Large-Scale Hierarchical Image Database," in *Proc. IEEE CVPR*, 2009, pp. 248–255. |
| [9] | S. Ioffe and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," in *Proc. ICML*, 2015, pp. 448–456. |
| [10] | N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," *J. Machine Learning Research*, vol. 15, pp. 1929–1958, 2014. |
| [11] | R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626. |
| [12] | M. D. Zeiler and R. Fergus, "Visualizing and Understanding Convolutional Networks," in *Proc. European Conf. Computer Vision (ECCV)*, 2014, pp. 818–833. |
| [13] | A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019. |
| [14] | F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *J. Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. |
| [15] | Streamlit. [Online]. Available: https://streamlit.io |


*Reference [1] and reference [2] are the project's own primary sources. Reference [1] has been verified against the primary publication as part of this report's base-paper verification pass (see Appendix A.6). Reference [2] awaits final licence/version confirmation before submission. References [3]–[15] are standard citations that should be checked against publisher records before submission.*

# Appendix A · Supporting Material

## A.1 Sample result-file excerpt

An abbreviated excerpt of *experiments/resnet152/results/final_results.json* (values rounded; this file holds the original 1,311-image evaluation):


```
{
  "backbone": "resnet152",
  "neural_network": { "best_epoch": 8, "val_accuracy": 98.25, "val_loss": 0.0597 },
  "deep_features": { "dimension": 128, "train_samples": 4569,
    "val_samples": 1143, "test_samples": 1311 },
  "pca": { "components": 64, "explained_variance": 0.9939 },
  "svm": { "kernel": "linear", "C": 1.0 },
  "validation": { "accuracy": 0.9825, "macro_f1": 0.9820 },
  "test": { "accuracy": 0.9840, "macro_f1": 0.9832,
    "confusion_matrix": [[292, 7, 0, 1], [2, 299, 3, 2],
      [0, 1, 404, 0], [3, 1, 1, 295]] }
}
```


## A.2 Duplicate-audit summary excerpt

Excerpt of *duplicate_leakage_audit.json* (created 20 September 2026):


```
{
  "status": "MATERIAL_DUPLICATION",
  "split_counts": { "train": 4569, "validation": 1143, "test": 1311 },
  "images_hashed": 7023,
  "cross_split_exact_or_pixel_duplicate_pairs": 464,
  "high_confidence_near_duplicate_pairs": 98,
  "near_duplicate_review_pairs": 481,
  "material_pairs": 562,
  "thresholds": { "candidate_phash_hamming_max": 4, "candidate_dhash_hamming_max": 6 }
}
```


## A.3 Primary corrected evaluation (material_clean, n = 1,061)

| Backbone | Accuracy | Macro precision | Macro recall | Macro F1 | Original accuracy |
|---|---|---|---|---|---|
| VGG16 | 94.816% | 94.134% | 94.583% | 94.302% | 94.966% |
| VGG19 | 93.025% | 92.249% | 92.734% | 92.415% | 93.516% |
| ResNet50 | 98.303% | 98.086% | 98.192% | 98.134% | 98.093% |
| ResNet152 | 98.398% | 98.236% | 98.402% | 98.315% | 98.398% |
| EfficientNet-B3 | 98.680% | 98.544% | 98.481% | 98.511% | 98.474% |

## A.4 Grad-CAM ratings for the 20-case protocol

ResNet152, seed 42, five images per class, opacity 0.55. "Best" is the map judged most anatomically relevant. Ratings are manual and subjective. Cases marked † were later identified as material duplicates.

| # | Class | Image | Layer 3 | Layer 4 | Fused | Best |
|---|---|---|---|---|---|---|
| 1 | glioma | Te-glTr_0002 | Partial | Poor | Poor | Layer 3 |
| 2 | glioma | Te-gl_0030 | Poor | Poor | Poor | Layer 4 |
| 3 | glioma | Te-gl_0085 | Poor | Poor | Poor | Layer 3 |
| 4 | glioma | Te-gl_0148 | Partial | Poor | Poor | Layer 3 |
| 5 | glioma | Te-gl_0282 | Poor | Partial | Partial | Layer 4 |
| 6 | meningioma | Te-me_0012 † | Good | Good | Good | Layer 4 |

| # | Class | Image | Layer 3 | Layer 4 | Fused | Best |
|---|---|---|---|---|---|---|
| 7 | meningioma | Te-me_0118 | Partial | Poor | Poor | Layer 3 |
| 8 | meningioma | Te-me_0137 | Partial | Good | Good | Layer 4 |
| 9 | meningioma | Te-me_0155 | Partial | Partial | Partial | Layer 4 |
| 10 | meningioma | Te-me_0204 | Partial | Good | Partial | Layer 4 |
| 11 | notumor | Te-no_0060 † | N/A | N/A | N/A | — |
| 12 | notumor | Te-no_0157 | N/A | N/A | N/A | — |
| 13 | notumor | Te-no_0191 | N/A | N/A | N/A | — |
| 14 | notumor | Te-no_0216 | N/A | N/A | N/A | — |
| 15 | notumor | Te-no_0352 | N/A | N/A | N/A | — |
| 16 | pituitary | Te-piTr_0008 | Partial | Poor | Partial | Layer 3 |
| 17 | pituitary | Te-pi_0010 | Poor | Poor | Poor | Layer 3 |
| 18 | pituitary | Te-pi_0078 | Poor | Poor | Poor | Layer 3 |
| 19 | pituitary | Te-pi_0267 | Partial | Poor | Partial | Layer 3 |
| 20 | pituitary | Te-pi_0282 | Partial | Poor | Partial | Layer 3 |

*Notes: Te-no_0191 produced a constant (blank) map for one layer; Te-no_0216 was flagged for expert review as a possible label anomaly and not relabeled; Te-pi_0078 was misclassified as glioma by both heads.*

## A.5 Repository structure

```
brain_tumor_project/
  model_cnn_lstm.py     generic CNN–LSTM model; SUPPORTED_BACKBONES
  app.py, gui/         Streamlit application and support modules
  src/              data_pipeline_torch.py, save_split.py, check_dataset.py
  actual_training_scripts/  train_backbone.py, evaluate_backbone.py, diagnostics
  experiments/<backbone>/   models/, features/, pca/, results/ (five backbones)
  data/splits/       train_indices.npy, val_indices.npy
  gradcam_evaluation/    Grad-CAM ratings CSV and figures (ResNet152)
  occlusion_evaluation/   occlusion comparison CSV (ResNet152)
  final_project_qc.py     duplicate / leakage audit
  evaluate_leakage_clean_test.py  post-hoc leakage-corrected evaluation
  final_project_docs/     audit outputs, corrected evaluation, environment
  models/, pca_features/    legacy root-level ResNet152 artifacts (superseded)
```

## A.6 Base-paper reproduction traceability matrix

This matrix is added in the base-paper verification pass. Rows are the elements the base paper defines; columns state where each is implemented in the project, evidence file(s), and what differs. "Report §X" refers to this report; "[1, §X]" refers to Bérété et al. 2026.

| # | Base-paper element | Paper loc. | Implemented? | Project file / evidence | Difference / note |
|---|---|---|---|---|---|
| 1 | Four-class MRI classification (glioma / mening. / notumor / pituitary) | §3.1 | Yes | Report §1, §3.3 | Same task; identical class names and label ordering |
| 2 | Dataset: Kaggle Nickparvar Brain Tumor MRI (7,023; 5,712 + 1,311) | §3.1 | Yes | Report §1, §3.3; Appendix A.2 | Same dataset; audit revealed pre-existing cross-split duplication in the supplied partitions |
| 3 | Test-set class supports (300 / 306 / 405 / 300) | §3.1 | Yes | Report §3.3, §6.3 | Identical |
| 4 | Data augmentation (H/V flip; bright ±0.1; contrast 0.8–1.2; sat 0.8–1.2; hue ±0.1) | Alg. 1 | Yes (equivalent) | Report §4.1; src/data_pipeline_torch.py | torchvision ColorJitter(0.1, 0.2, 0.2, 0.1) yields identical ranges |
| 5 | Framework: TensorFlow | §3.2 | **No — deliberate** | Report §1, §5 | Migrated to PyTorch after RTX 50-series sm_120 PTX incompatibility; environmental, not scientific |
| 6 | Backbones: VGG16, VGG19, ResNet50, ResNet152, EfficientNetB3 | §3.2 | Yes, all five | Report §4.3; model_cnn_lstm SUPPORTED_BACKBONES | Same five; independent alternatives, not an ensemble |
| 7 | ImageNet pretraining of CNN backbones | §3.2 | Yes | Report §4.3 | Same (V1/V2 weight strings noted) |
| 8 | CNN → reshape → LSTM chain | §3.2, Fig. 2 | Yes | Report §3.2, §4.2 | Report clarifies LSTM sees spatial positions, not time — a valid rewording |
| 9 | LSTM with forget/input/output gates (Eq. 1–6) | §3.2 | Yes (nn.LSTM) | Report §4.2 | Hidden 256, batch-first, one layer |
| 10 | BatchNorm + Dropout + Dense after LSTM | §3.2, Fig. 2 | Yes | Report §3.2, §4.2 | BatchNorm1d → Dropout(0.3) → Linear(256→128) → ReLU |
| 11 | StandardScaler (zero mean, unit variance) | §3.2 | Yes | Report §4.5; evaluate_backbone.py | Fitted on training only, as the paper prescribes |
| 12 | PCA to min(n_samples, n_features, 64) → PCA(64) | §3.2 | Yes | Report §4.5 | Same; explained variance 99.18–99.45% per backbone |
| 13 | Linear SVM, C = 1.0 | §3.2 Eq. 7–8 | Yes | Report §4.5 | Same kernel and regularization |
| 14 | SVM *probability=True* | §3.2 | **No — differs** | Report §4.5 | GUI uses decision_function(); preserved legacy logs show True on some runs |
| 15 | Optimizer study: Adam + SGD + AdaGrad + RMSprop (5×4 = 20 runs) | §4.2, Tab. 2 | **Partial — RMSprop only** | Report §8.3, §9.3 (future work) | ~4× less compute than the paper's protocol |
| 16 | Learning rate 1e-4 | Tab. 2 | Yes | Report §4.4 | Same |
| 17 | Loss: categorical cross-entropy | Tab. 2 | Yes | Report §4.4 | Same |
| 18 | Batch size 32 | Tab. 2 | **No — differs** (8) | Report §4.4 | Small-batch training with drop_last=True for BatchNorm1d stability |
| 19 | Max epochs 30 (fixed) | Tab. 2 | **Partial** | Report §4.4, §6.1 | 10 default with early stopping; VGGs extended to 30 |
| 20 | Dropout 0.3 | Tab. 2 | Yes | Report §4.4 | Same |
| 21 | Freezing / fine-tuning policy | Not specified | Project-specific | Report §4.3 | VGG16/VGG19 frozen; others fine-tuned — asymmetry not in the paper |

| # | Base-paper element | Paper loc. | Implemented? | Project file / evidence | Difference / note |
|---|---|---|---|---|---|
| 22 | Metrics: accuracy, precision, recall, F1 | §4.1 | Yes | Report §6.2, §6.4 | Same; also per-class support and macro/weighted F1 |
| 23 | Confusion matrices (per optimizer × backbone) | Figs. 5–8 | Partial (RMSprop only) | Report §6.2, §6.4 | Report shows ResNet152 / EfficientNet-B3 corrected matrices; paper shows 20 × 4 × 5 |
| 24 | Comparative table against other SoTA | Tab. 4 | For context only | Report §8.4 | Report does not claim to outperform paper's cited SoTA |
| 25 | Best result (ResNet152+LSTM+SVM RMSprop = 99% acc/F1) | Abstract, Tab. 3, Tab. 4 | **Not reproduced at same headline** | Report §6.2, §6.4, §6.5 | ResNet152 = 98.40% (orig) / 98.398% (corr); ~0.6 pp below paper; VGG19 ~3 pp below paper |
| 26 | Grad-CAM / other XAI | Absent (future work in §4.4) | **Yes — project extension** | Report §4.6, §6.6 | Grad-CAM on the neural head at ResNet layer 3, 4, and fused |
| 27 | Occlusion sensitivity on the full pipeline (final SVM) | Absent | **Yes — project extension** | Report §4.6, §6.6 | Interrogates the deployed classifier where Grad-CAM cannot |
| 28 | Streamlit / interactive research GUI | Absent | **Yes — project extension** | Report §4.8; app.py, gui/* | Four-tab research application |
| 29 | Cross-split duplicate / near-duplicate audit | Absent | **Yes — project extension** | Report §4.7, §6.3 | Hashing + pHash + dHash + MAE/correlation across three partition pairs |
| 30 | Leakage-corrected evaluation (post-hoc, no retraining) | Absent | **Yes — project extension** | Report §3.5, §6.4 | Primary policy material_clean (n = 1,061); two sensitivity policies |
| 31 | Statistical rigour (multiple seeds, CIs, paired tests) | Absent | Not implemented | Report §7.2, §9.3 | Both project and paper are single-seed / single-split |
| 32 | Ablations (no-LSTM, no-PCA, alternative orderings) | Absent | Not implemented | Report §7.2, §9.3 | Future work in both |
| 33 | External-dataset / patient-level validation | §4.4 (as future work) | Not implemented | Report §7.2, §9.3 | Future work in both |

## A.7 Final assessment (viva-defensible)

This section answers eight questions a viva panel is likely to ask. The answers are evidence-based and follow the instruction not to flatter the project.

### A.7.1 Did we successfully implement the core architecture proposed by the base paper?

**Yes.** All five backbones are implemented in the same CNN → spatial-sequence → LSTM → BatchNorm → Dropout → Dense(128) → StandardScaler → PCA(64) → linear SVM pipeline the paper describes (traceability rows 6–13). The five pipelines run end-to-end, produce metrics, and drive the Streamlit application. This part of the reproduction is complete.

### A.7.2 How closely did we reproduce the methodology?

**Partially, with disclosed deviations.** The reproduction is faithful in architecture, loss, learning rate, dropout, PCA size, and SVM kernel/C. It diverges deliberately in:

- **Framework** — PyTorch instead of TensorFlow (RTX 50-series sm_120 incompatibility; environmental, not scientific).
- **Optimizer scope** — RMSprop only, not the paper's 5 × 4 matrix.
- **Batch size** — 8, not 32 (drop_last=True required for BatchNorm1d).
- **Epoch budget** — 10 default with early stopping, VGGs extended to 30; the paper uses 30 fixed.
- **VGG freezing** — the paper describes no freezing; the project froze VGG16/VGG19 after training collapse and fine-tuned the other three.
- **SVM probability flag** — paper enables it; the project uses *probability=False*, with the GUI using *decision_function()*.

None of these deviations is hidden: each is documented in §4 and §8. The strongest single-line honest description is **"paper-inspired partial reproduction, not an exact reproduction"**.

### A.7.3 How closely did our results reproduce the paper's reported performance?

The paper's headline is **ResNet152+LSTM+SVM (RMSprop) = 99% accuracy / 99% F1** [1, Table 3]. Under RMSprop the paper reports four backbones at 99% and EfficientNetB3 at 98%. The project's results are:

| Backbone | Paper acc (RMSprop) % | Project original test acc % | Project material_clean % |
|---|---|---|---|
| VGG16 | 94 | 94.97 | 94.816 |
| VGG19 | 96 | 93.52 | 93.025 |
| ResNet50 | 99 | 98.09 | 98.303 |
| ResNet152 | 99 | 98.40 | 98.398 |
| EfficientNet-B3 | 98 | 98.47 | 98.680 |

**Interpretation.** VGG16 and EfficientNet-B3 land within ≈1 pp of the paper. ResNet50 and ResNet152 land ≈0.6–0.9 pp below the paper's 99%. VGG19 lands ≈3 pp below the paper — the largest gap. Near-identical numbers for three backbones and one meaningful VGG19 gap is a **credible partial reproduction**, not a headline-matching one. The report does not claim to replicate the 99% figure.

### A.7.4 What important differences exist?

- **Freezing asymmetry.** The five-backbone comparison is not "architecture only"; VGGs are fixed feature extractors with a trained read-out while the other three adapt their convolutional weights (report §4.3).

• **Data-integrity finding.** The project audit found pre-existing cross-split duplication in the paper's own dataset partition (562 material pairs). This affects anyone who uses those directories as-is — including the base paper, whose results are reported without any duplicate check. Arguably the single most consequential difference.

• **Explainability and integrity integrated into the study.** The base paper's §4.4 treats XAI and larger validation as future work; the project performs them at small scale with disclosed caveats and folds their findings back into the interpretation (§6.6, §6.7). A methodological difference in kind, not just scope.

### A.7.5 What did we add that the base paper did not contain?

Concretely (traceability rows 26–30): Grad-CAM on the differentiable neural head; full-pipeline final-SVM occlusion sensitivity with prediction-flip counts and normalization caveats; cross-split duplicate/near-duplicate audit using five complementary measures; post-hoc leakage-corrected evaluation under three predefined policies; and a Streamlit research application loading standardized per-backbone artifacts. None of the above appears in the base paper.

### A.7.6 Are Grad-CAM and final-SVM occlusion a defensible mini-project-level extension?

**Yes, at the mini-project / undergraduate research level.** Grad-CAM is a well-established method [11], and its use on hybrid CNN-based classifiers is standard. Where the project earns its credibility is in *not* overclaiming: (i) Grad-CAM explains the neural head, not the deployed SVM; (ii) occlusion is added because Grad-CAM cannot explain the SVM; (iii) occlusion maps are normalized per image and flip counts are reported alongside the maps. The single-rater, 20-case protocol is a legitimate weakness (§6.6, §7.2) — but it is stated openly. If a reviewer objects that occlusion is not a novel method, the correct reply is that novelty was never claimed; the extension is a **methodological** one — matching each explainer to the classifier it can actually explain — not an algorithmic one.

### A.7.7 Does the leakage audit materially change the project conclusion?

**Yes for interpretation; no for the numeric ranking.**

• For **interpretation**: it downgrades the headline claim. Before the audit, ~98% on a public benchmark would be presented as a strong within-dataset result. After the audit, the honest statement is ~98% on a test set that shared pre-existing image-level content with the training partitions, and remained ~98% on the 1,061-image material-clean subset. Weaker but truer.

• For the **numeric ranking**: the post-hoc corrected accuracies moved by less than 0.5 pp for every backbone, and the ordering (three residual/EfficientNet models above the two frozen VGG models) was preserved. The audit does not overturn the comparison.

• For **future work**: the audit produces a specific, non-speculative next step — duplicate-group-aware retraining — which the project defers rather than pretends to have done. Disclosed in §6.4 and §9.3.

### A.7.8 What is the strongest defensible contribution of the completed project?

**Not the accuracy number.** The strongest, most defensible contribution is the **methodological triangulation** the project demonstrates on a public MRI benchmark: *performance* × *explainability* × *dataset integrity* × *reproduction discipline, examined together on the same trained pipelines.* Concretely:

• A standardized, artifact-preserving five-backbone comparison that made the leakage audit and corrected re-scoring possible *without* retraining (§6.4 explicitly relies on the saved test-PCA features and SVMs).

• Two explanation paths, each matched to the classifier it can actually explain, with quantified normalization/effect-size caveats — a discipline the base paper does not exercise.

• A duplicate audit that identified pre-existing cross-split contamination in a widely used public dataset and quantified its effect. Whether or not any reviewer accepts the specific pHash/dHash thresholds, the finding is reproducible and independently checkable.

• A traceable base-paper reproduction (Appendix A.6) that states row-by-row what was implemented, what was deviated from, and what was extended.

That combination — not the ≈98% — is what a viva committee should be directed toward.

*End of report.*